



Full-length article

Optimization and design of hybrid power system using HOMER pro and integrated CRITIC-PROMETHEE II approaches

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ARTICLE INFO

Keywords:

Human excreta
Solar PV
Biogas
CRITIC
PROMETHEE II
HOMER

ABSTRACT

Interrupted power supply and poor access to electricity (15%) have been persistent problems in Malawi for decades. Diversification of resources is required to solve the challenges. Therefore, this study is aimed at conducting a techno-economic analysis of a hybrid system to ensure electricity reliability, bill reduction, and reduced grid demand at a school. The study investigated six hybrid system scenarios that had various combinations of the grid, diesel, solar PV, wind, biogas, and battery. The HOMER Pro model was used to determine the best system combination. Six scenarios were examined further using the CRITIC-PROMETHEE II approaches. The best system configuration included grid, solar PV, and biogas electricity. Biogas generation is mainly from human excreta. The study also investigated the effect of the 2022 inflation rate on the financial performance of the system, which had shown a sharp increase in capital, replacement, O&M costs, and payback time. The proposed system's levelized cost of electricity is 0.095 \$/kWh, which is less than Malawi's grid's levelized cost of 0.11 \$/kWh. Environmentally, the system could help to reduce greenhouse emissions, including those from the sewage system. Therefore, schools and governments should invest in alternative energy generation.

1. Introduction

Growing energy demand due to population growth and urbanization put pressure on the national electricity grid. For example, by 2050, 60% of Africans are expected to live in cities, resulting in increased electricity demand [1]. Furthermore, most rural areas have no access to electricity, which harms electricity-dependent developments such as telecommunications, health care, and education [2]. Effort on improving energy access through sustainable development goal 7 (SDG7) has shown progress in increasing electricity access in several countries such as Kenya, Bangladesh, and India [3]. Thus, governments must put more effort into increasing access through the right policies, proper planning, regulations, and incentives and decentralized renewables must be encouraged.

Despite this, many countries still rely heavily on fossil fuels. Sadly, the price spike brought on by the war in Ukraine and the sanctions against Russia have proven that fossil fuels are unreliable [4]. Fossil fuels are used in most sectors, including manufacturing and transportation. This has had a serious adverse effect on many nations, including

developing nations [3]. The cost of living and the inflation rate have both significantly increased around the world. For instance, Malawi's inflation rate was 9.34% in 2021, but the conflict in Ukraine caused it to rise to 24.6%. subsequently influencing a variety of financial projects.

Malawi's population is about 19 million [5] and the country has an electrification rate of 15% [6] with about 12% connected to the grid [7]. Malawi has significant renewable energy resources, which include rivers, solar, wind, geothermal, and biomass. In 2022, the installed capacity was about 546 MW [8] with heavy dependence on biomass for cooking [9] and small-scale solar PV for phone charging, irrigation, and lighting [10]. Hydropower dominates power generation followed by new solar PV with capacities of 60 MW and 20 MW in Salima and Dedza respectively [11]. Dropping water levels reduced generation capacity, resulting in frequent power outages [12]. However, as of 2020, Malawi required over 700 MW of electricity to meet the existing demand [13]. Thus, the rising energy demand, combined with limited electricity investment, renders grid electricity

Abbreviations: HOMER, Optimization of Multiple Energy Resources; NPC, Net present cost; LCOE, Levelized Cost of Energy; LCOH, Levelized cost of hydrogen; PROMETHEE, Preference Ranking Organization Method; CRITIC, Criteria Importance Through Intercriteria Correlation; TS, Total solid; TD, Total discharge; TSFD, Total solid of fresh discharge; BP, Biogas production rate per kg.TS; HRT, Hydraulic retention time; Q, Daily total effluent added to a digester; V, Total volume of a digester; PV, Solarphotovoltaic; BG, Biogas generator; DG, Diesel generator; BT, Battery; HG, Hydrogen gas; WT, Wind turbine; MCDM, Multiple-criteria decision-making

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<https://doi.org/10.1016/j.grets.2022.100005>

Received 8 October 2022; Received in revised form 22 November 2022; Accepted 27 November 2022

Available online 30 November 2022

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unreliable and insufficient [14]. Frequent power outages have an impact on businesses that rely on electricity, including schools, which are highly dependent on electricity for most of their activities, such as laboratories, computers, and general studies at night. Renewable energy sources are critical to overcoming these obstacles and ensuring the development of sustainable energy planning [15,16].

Utilizing human waste can be one of the alternatives to overcome these electricity challenges. Biogas is commonly produced from agricultural waste, food waste, or cattle manure. However, using human waste, called human excreta, has advantages in terms of process, environment, health, and economic impact [17]. Septic tanks converted to biogas digesters have many advantages including cost-effectiveness, production of safe fertilizer, and fight against global warming [18]. Human excreta have a sanitation advantage and do not require any additional starter (microorganisms seed) during production [19]. Mudassar & Kim [20] reported that $0.35 \text{ m}^3/\text{kg}$ of biogas can be generated from 35 kg of human excreta, resulting in 26.8 kWh of energy. Some African and Asian countries are already using human excreta in biogas production [21].

Recent years have seen many technical and economic studies on the use of hybrid renewable energy resources. Many hybrids renewable energy system design and optimization tools have been developed but the Hybrid Optimization of Multiple Energy Resources (HOMER) modeling tool has emerged as one of the best and most popular tools globally [22, 23]. Several authors have demonstrated that hybrid renewable energy systems can complement each other to ensure a reliable electricity supply. Khalil et al. [24] investigated a hybrid system comprised by wind and solar PV to reduce bills for the Baluchistan's Seashore. The study utilized the minimization of net present cost (NPC) and reported a 64% reduction in pollution gas emissions. Nebey [25] studied a hybrid system consisting of solar PV, wind, and hydropower for a community load in Ethiopia. The study utilized low NPC to choose the best combination using HOMER software. Ghaffari & Askarzadeh [26] optimized a hybrid system composed of solar PV, diesel generator, and fuel cell. The study utilized the minimization of NPC to determine the best combination. Riyatasyah et al. [27] investigated a grid-connected system comprised of solar PV, wind, and battery for Syiah Kuala University, Indonesia. The system was optimized using HOMER software by minimizing NPC and Levelized Cost of Energy (LCOE). According to the study, the system can contribute up to 82% of the required electricity. Jumare et al. [28] assessed a decentralized grid-connected solar PV, wind, and biogas power system in Nigeria. The minimization of NPC and LCOE helped in determining the best systems combination. The study further reported an increase in energy supply by 3% and a reduction of LCOE by 81%. Khan et al. [29] carried out a techno-economic assessment of a standalone solar/ wind hybrid system for a rural area in India. The study indicated that the LCOE for the optimized system was 0.192 \$/kWh. In Turkey, Akarsu et al. [30] used HOMER to optimize the hybrid system comprised of solar, wind, diesel generator, battery, and hydrogen storage. It was found that the hybrid system had an LCOE of 0.376 \$/kWh. In India, Singh et al. [31] optimized a hybrid system comprising solar, fuel cell and biomass was optimized using HOMER Pro Software. The system's LCOE was found to be 15.064 Rs/kWh. In Ghana, Ampah et al. [32] optimized a hybrid system comprising solar, wind, and biomass considering the fuel cell electric vehicles. HOMER Pro and MCDM approaches were utilized and from the results, the LCOE and LCOH were \$0.52/kWh and \$9.09/kg respectively.

Several optimization studies employ NPC and LCOE. However, during the optimization process, other economic, environmental, and other factors are not considered. Thus, multicriteria decision-making (MCDM) approaches are increasingly being used to incorporate other factors when selecting energy options. MCDM entails a series of evaluations of a limited number of options using a limited set of specified criteria [33]. This study employs the multi-criteria decision-making techniques of the Preference Ranking Organization Method (PROMETHEE) and Criteria Importance Through Intercriteria Correlation (CRITIC). The CRITIC Method's weights consider the degree of

contrast between the alternatives as well as the evaluation criteria's inherently inconsistent nature [34]. Several other studies have used CRITIC alone or combined with other methods to determine the best choice. For example, choosing the best food waste treatment procedure [35], estimating sustainable supply chain risk management [36], evaluating the effectiveness of VOC-contaminated soil [37], assessing the operation safety of urban rail transit [38], ranking livestock feeding stuff, and assessing the risk management of subsea pipelines [39]. The PROMETHEE II method is a user-friendly outranking approach that is effective in real-world planning situations and complete in the ranking [40]. It considers the imprecision and uncertainty associated with evaluating the decision criteria [41]. Several studies have applied this method in several applications. For example, assessing irrigation methods [42], equipment selection [43], selecting hazardous waste disposal enterprises [44], and selecting solar power plant locations [45].

As evidenced in the previous studies, optimizing a hybrid renewable energy system is a complex issue because it is site-specific as it depends on location, economic and environmental parameters, load demand, and available renewable energy resources. Most optimal systems have been chosen based on LCOE, NPC, and minimum cost of generation. However, decision-making in renewable energy technologies (RET) is complex as such a few studies have incorporated both economic aspects and MCDM. Furthermore, there are limited studies considering human excreta for biogas generation with consideration of a school for a large collection of human excreta. Therefore, in this study, to select the most optimal hybrid energy system, the economic aspect, environmental aspect, and biogas from human excreta have been considered using MCDM (CRITIC and PROMETHEE II). Table 1 gives a summary of some of the studies conducted in several locations using different approaches.

Given the state of Malawi's energy sector and the recent interest in renewable energy generation, few studies have assessed the potential of biogas from human excreta for hybrid system optimization. Therefore, this paper is aimed at conducting a techno-economic analysis of a hybrid system to ensure electricity reliability, bill reduction, and reduced grid demand at a school. The following are the main contributions of this paper:

- A novel grid-connected hybrid renewable energy system was designed to solve electricity challenges and reduce electricity bills at a school.
- Sizing a grid-connected hybrid system comprised of solar PV and biogas from human excreta at a school setting.
- Determining the optimal hybrid system by considering economic, technical, and environmental aspects using MCDM (CRITIC and PROMETHEE II) approaches.
- The use of human excreta would also offer sanitation and economic management of municipal waste at a school.

The remaining parts of this paper are organized as follows: Section 2: Materials and methods. Section 3: Results and discussion, and Section 4: Conclusion.

2. Materials and methods

2.1. Study location

The research was carried out at Blantyre Secondary School, which is in the southern part of Malawi, in the city of Blantyre, at a latitude of $15^{\circ}47.09' \text{ S}$ and a longitude of $35^{\circ}1.77' \text{ E}$. Fig. 1 depicts the chosen site's location. The country has over 527 secondary schools with over 44 government-supported boarding secondary schools [58]. In this study, Blantyre Secondary School (BSS) was chosen as a case study. This is because BSS is one of the big government secondary schools located within the city of Blantyre and it can represent all government secondary schools in Malawi. The school had a total number of students of about 480.

Table 1
Summary of literature review in different locations.

Research	Year	Location	Mode	System	Human excreta biogas	HOMER tool	MCDM approach
Dehshiri [46]	2022	Iran	Standalone	FC, HT, WT, PV, DG, BT, EL	No	Yes	Yes
Odoi-Yorke et al. [47]	2022	Ghana	Off-grid	PV, DG, WT, NG, BT	No	Yes	Yes
Roy et al. [48]	2022	India	Off-grid	WT, PV, BG, FC	No	No	No
Elkadeem et al. [49]	2021	Kenya	Off-grid	PV, WT, DG, BT	No	Yes	Yes
Ukoba et al. [50]	2020	Nigeria	Off-grid	PV, WT, BG, BT	No	Yes	Yes
Ribó-Pérez et al. [51]	2021	Zambia	On-grid	PV, BG, Grid, BT	No	Yes	No
Li et al. [52]	2022	China	Off-grid, on-grid	PV, WT, BG, Grid	No	Yes	No
Chowdhury et al. [53]	2022	Bangladesh	Standalone	PV, WT, BG, BT, DG	No	Yes	Yes
Ampah, Jin, et al. [54]	2022	China	Standalone	PV, WT, BG, BT	No	Yes	No
Chisale & Mangani [55]	2021	Malawi	On-grid	PV, BT, Grid	No	Yes	No
Rad et al. [56]	2020	Iran	On-grid, off-grid	PV, WT, BG, FC, BT, HT	No	Yes	Yes
Podder et al. [57]	2022	Bangladesh	Off-grid	PV, BG, BT	No	Yes	No

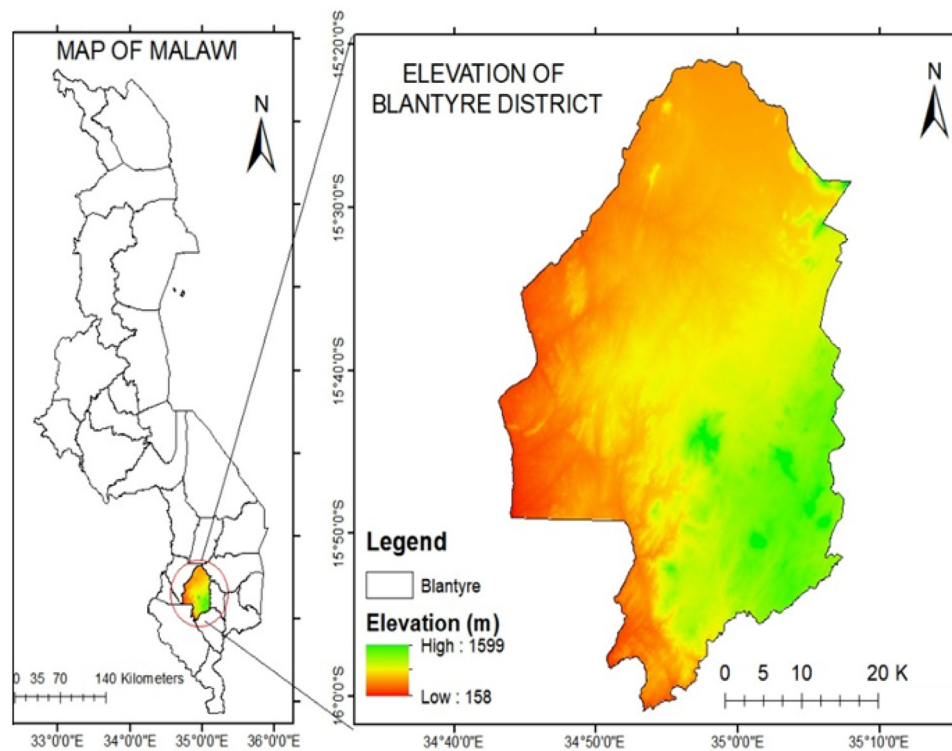


Fig. 1. Location of study location in Malawi.

2.2. Renewable energy resource assessment and load demand

The meteorological data for the school was obtained from NASA's surface meteorology and solar energy database for 22 years. Additionally, wind speed data was also obtained from the NASA database at an anemometer height of 50 m above the surface of the earth for terrain like airports for 10 years. Table 2 shows meteorological data for the school. The annual average daily radiation, wind speed, and temperature are 5.38 kWh/m²/day, 4.81 m/s, and 23.33 °C respectively.

2.3. Load estimation

Blantyre Secondary School mainly uses its power in computers, lighting, and other applications such as printers and small laboratory equipment. The school has hostels for both boys and girls, a cafeteria, a hall, a resource center, classrooms, and an administration building. Table 3 shows the summary of the quantity and power ratings of available appliances. A large amount of electricity is consumed in lighting with about 66% of the total load. Second, computers in offices

and laboratories consume approximately 30% of the total load. The school had scaled peak load and annual average demand of 23.7060 kW and 334.750 kWh/day respectively with a load factor of 0.5884. Fig. 2 shows the daily load profile for the school.

2.4. System design

The investigation's first step is to assess renewable energy resources. The second step involves estimating the school's electricity load. Then, using several input parameters and available renewable energy resources, HOMER software was used to determine the energy mix and other financial output. Fig. 3. depicts the overall methodology for the pre-feasibility study of Blantyre Secondary School's energy mix. The design was created in HOMER software, considering local conditions, cost parameters, energy resources, and other important simulation parameters. The optimization in HOMER was simulated for one year using hourly steps to meet the power demand with the lowest NPC. Several combinations of the grid, solar PV, wind, biogas, and battery were proposed. Thus, six combinations defined as scenarios 1–6 were simulated as depicted in Fig. 4.

Table 2
Meteorological data for Blantyre School.

Month	Clearness index	Daily radiation (kWh/m ²)	Temperature (°C)	Wind speed (m/s)
January	0.468	5.32	24.1	4.21
February	0.493	5.46	23.87	4.03
March	0.527	5.43	23.52	4.28
April	0.57	5.19	22.56	4.1
May	0.61	4.84	21.24	4.33
June	0.594	4.35	19.8	4.55
July	0.591	4.48	19.98	4.95
August	0.614	5.26	21.96	5.11
September	0.634	6.21	25.18	5.58
October	0.604	6.49	26.55	5.84
November	0.544	6.12	26.38	5.89
December	0.476	5.43	24.87	4.85

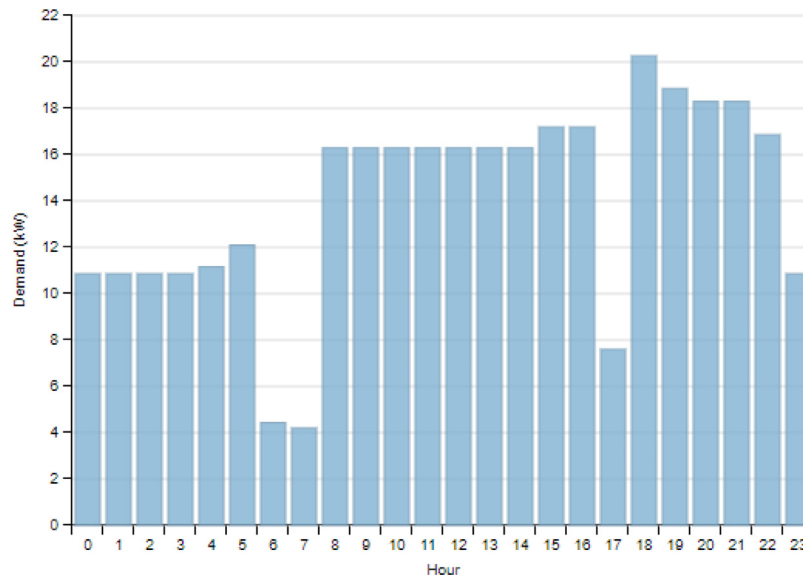


Fig. 2. Daily load profile for the school.

Table 3
Summary of equipment and their power ratings.

Equipment	Quantity	Rated power (W)	Connected load (kW)
Computers	53	200	10.6
Security lights	53	75	3.975
	1	60	0.06
Inside lights	2	400	0.8
	5	50	0.25
	231	75	17.325
Internet router	4	15	0.06

2.5. System components of the model

2.5.1. Grid electricity

Blantyre Secondary School relies on grid electricity for its operations. However, grid electricity in Malawi is currently not reliable. Available statistics indicate that the number of power outages per month was 7.4 while the average duration of the outage was 3.6 h [59]. In HOMER, mean outage frequency and mean repair time was inputted as 88.8 per year and 3.6 h respectively to calculate random grid outages in a year. According to Malawi Energy Regulatory Authority [60], the cost of electricity on the grid for this application as of February 2022 was 0.11 \$/kWh hence entered in HOMER.

2.5.2. Solar PV

The cost of a Bluesun solar panel, which is a Monocrystalline Silicon in China, is 0.26 \$/watts (260 \$/kW) [61]. However, according to the

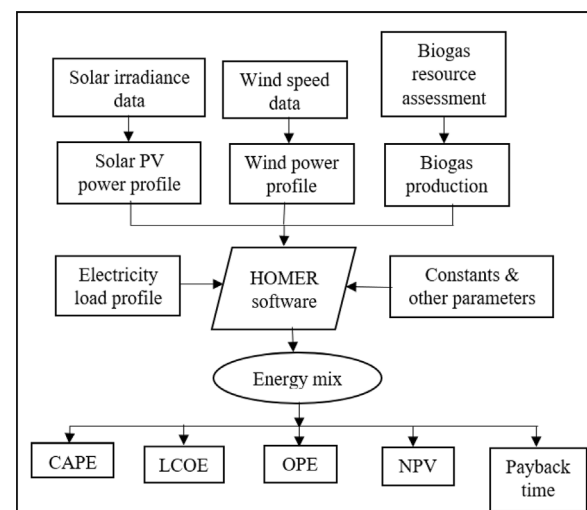


Fig. 3. Overall methodology flow diagram.

cost of solar panels in Malawi, 1 kW costs about \$950. Because solar PV technology has not yet reached maturity, the replacement cost was assumed to be \$800, implying that its cost will most likely decrease in the future. The cost of operation and maintenance was estimated to be \$1 per kW per year. 25 years has been considered as the lifetime

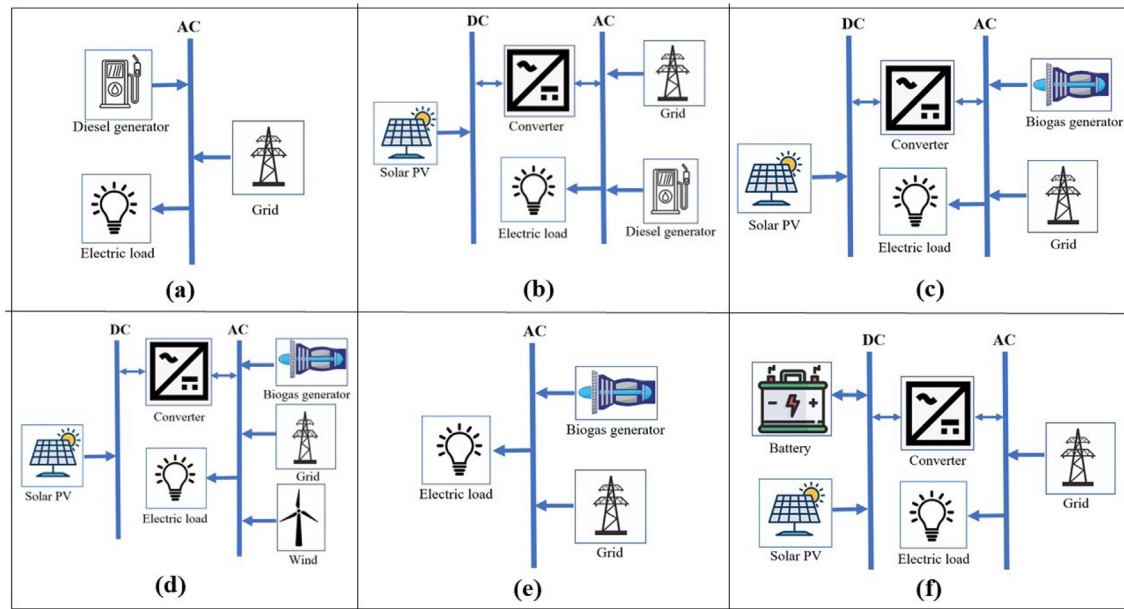


Fig. 4. Schematic diagrams for six scenarios (a) Scenario 1, (b) Scenario 2, (c) Scenario 3, (d) Scenario 4, (e) Scenario 5, (f) Scenario 6.

of solar modules with a derating factor of 90%. Eq. (1) was used to compute solar PV power generation.

$$P_{PV\text{ poweroutput}} = P_{pv,STC} f_{PV} \frac{G_T}{G_{T,STC}} [1 + K_P (T_C - T_{STC})] \quad (1)$$

Where $P_{pv,STC}$: Photovoltaic array at peak power (kWp).

f_{PV} : Derating factor (%)

G_T : Solar irradiance striking the PV array (kW/m²)

$G_{T,STC}$: Solar irradiance under standard test conditions (1 kW/m²)

K_P : Temperature coefficient of power (%/°C)

T_C : Photovoltaic temperature (°C)

T_{STC} : Ambient temperature (25 °C)

2.5.3. Biogas

The collection of students at a boarding school gives it good potential for the production of biogas. The study considered that an average human being of 50 kg mass can give average excreta of 0.5 kg. The recommended total solid (TS) contained in a material is 8%. The daily total discharge (TD) for the school can be computed using Eq. (2) [62]:

$$TD = \text{Discharge per student(kg)} \times \text{Number of students} \quad (2)$$

The TS value of fresh discharge for human excreta is 20%. Then the total solid of fresh discharge (TSFD) can be found using Eq. (3) below [62]:

$$TSFD = TD \times TS\text{value of fresh discharge (\% by wt.)} \quad (3)$$

Considering that the recommended total solid (TS) contained in a material is 8%. Therefore, using a simple proportion of 8% of effluent equal to the total solid, then, the total effluent required can be calculated. Thus, the amount of water to be added to the effluent can be calculated using Eq. (4) below:

$$\text{Added water} = \text{Total effluent} - \text{Daily total discharge} \quad (4)$$

The Biogas production rate per kg.TS of human excreta is 47.8%. The amount of gas produced per day (in m³) can be calculated using Eq. (5) below [63]:

$$\text{Gas} = TS_V \times TD \times BP \quad (5)$$

Where:

TS_V = TS value of fresh discharge(% by wt)

TD = Total discharge(kg/day)

BP = Biogas production rate per kg.TS.

Considering the hydraulic retention time (HRT) of 40 days and mesophilic temperature of 30 °C. Then, the total volume of a digester, V , can be calculated using Eq. (6) below [64]:

$$V = HRT \times Q \quad (6)$$

Where:

Q is the daily total effluent added to a digester.

2.5.4. Biogas generator

A 20 kW biogas generator costs about \$8,000 [65] in China. However, due to shipping, transportation through Tanzania, taxes, and installation costs, the cost of a generator was set at \$500 per kW for design purposes. The cost of replacing the generator was set at \$400, with the cost of operation and maintenance set at 0.03 \$/h. The rate of fuel consumption in relation to power output was calculated using Eq. (7).

$$F(t) = F_0 P_{rated} + F_1 P_{gen}(t) \quad (7)$$

where $F(t)$: The fuel consumption rate of diesel generator.

F_0 : Intercept coefficient

F_1 : Slope of the fuel curve

P_{rated} : Rated output power of diesel generator.

$P_{gen}(t)$: Real power output power of diesel generator.

2.5.5. Wind turbine

In Iran, Baneshi et al. [66] considered a wind turbine cost of 2,000 \$/kW. In China, a 3-kW wind turbine cost \$3,000 in China [67]. Due to geographical location, logistical, and installation challenges associated with a 3-kW wind turbine was set at \$6,360 (2120 \$/kW) while replacement and annual O&M cost was set at \$5,088 and \$127.2 respectively. The hub height was 12 m and the lifetime of the turbine is 20 years.

2.5.6. Charge controller and battery

The system requires a charge controller to ensure power regulation to and from the battery. It senses the battery is either fully charged or requires a recharge. This protects the battery from being damaged. When batteries are fully charged, the charge controller diverts excess

power to dump-load. This prevents the battery from being overcharged. The number of batteries required for the system can be calculated using Eq. (8).

$$N_{bat} = \frac{E_d \cdot n_d}{V_{bat} \cdot Ah \cdot DOD} \quad (8)$$

where n_d : The days of autonomy

E_d : The energy demand per day

V_{bat} : The voltage score of the battery

Ah: The hour score of amperes

DOD: The battery system depth of discharge

2.5.7. Power converter

The designed system is comprised of DC and AC components including a grid, generator, battery, and solar PV making the converter a useful component in a hybrid system. The converter capacity for the system is calculated using the Eq. (9).

$$P_{inv} = \frac{E_{L,max}}{\eta_{dc/ac}} \quad (9)$$

Where P_{inv} : the capacity of a power converter

$E_{L,max}$: the maximum energy demand required by the load (Wh)

$\eta_{dc/ac}$: conversion efficiency

2.6. CRITIC-PROMETHEE II approach

2.6.1. CRITIC approach

The CRITIC approach is one of the MCDM methods. The CRITIC approach was utilized to calculate the weights of criteria for the HOMER output hybrid system combinations. These combinations were categorized as Scenarios 1–6 defined in Fig. 4. The following were the steps utilized to determine the objective weights [32,68–70]:

Consider a decision matrix $X = [x_{ij}]_{m \times n}$, consisting of m alternatives and n criteria was constructed where $i = 1, \dots, m$ and $j = 1, \dots, n$, whereas x_{ij} represents the performance of measure of the i th for the j th criterion. Step 1 involves creating a normalized decision matrix using Eqs. (2) and (10). Step 2 involves calculating the standard deviation (σ_j) for the j th criterion from the normalized decision matrix. Step 3 involves determining a symmetric matrix of $n \times n$ using Eq. (12). Step 4 involves calculating a measure of conflict for each criterion using Eq. (13). Step 5 involves multiplying the standard deviation obtained in step 2 and the measure of conflict obtained in step 4. Step 5 is illustrated using Eq. (14). Finally, step 6 is to determine the final weights using Eq. (15).

$$r_{ij} = \frac{x_{ij} - \min(x_j)}{\max(x_j) - \min(x_j)} \quad (10)$$

$$r_{ij} = \frac{\max(x_j) - x_{ij}}{\max(x_j) - \min(x_j)} \quad (11)$$

$$\rho_{jk} = \frac{\sum_{i=1}^m (r_{ij} - \bar{r}_j)(r_{ik} - \bar{r}_k)}{\sqrt{\sum_{i=1}^m (r_{ij} - \bar{r}_j)^2 \sum_{i=1}^m (r_{ik} - \bar{r}_k)^2}} \quad (12)$$

$$Conflict = \sum_{k=1}^n (1 - \rho_{jk}) \quad (13)$$

$$C_j = \sigma_j \sum_{k=1}^n (1 - \rho_{jk}) \quad (14)$$

$$w_j = \frac{C_j}{\sum_{k=1}^n C_k} \quad (15)$$

2.6.2. PROMETHEE II approach

Consider decision challenge with m alternatives defined as A_i , for $i = 1, \dots, m$, and n criteria defined as C_j , for $j = 1, \dots, n$. Let the sets of $I = \{1, \dots, m\}$ and $J = \{1, \dots, n\}$ be the alternative and criterion indices, respectively. Thus, an alternative-criteria evaluation matrix is defined as $R = [r_{ij}]_{m \times n}$, where r_{ij} represents the evaluation of i th

alternative on j th criteria. The following steps were utilized [32,71–73]:

Step 1: construct a normalized matrix R using linear normalization using Eq. (16).

$$R_{norm} = [\tilde{r}_{ij}]_{m \times n} \quad (16)$$

Where,

$$\tilde{r}_{ij} = \begin{cases} \frac{r_{ij} - \min_{i \in I} \{r_{ij}\}}{\max_{i \in I} \{r_{ij}\} - \min_{i \in I} \{r_{ij}\}} & \text{for beneficial criterion } j\text{th} \\ \frac{\max_{i \in I} \{r_{ij}\} - r_{ij}}{\max_{i \in I} \{r_{ij}\} - \min_{i \in I} \{r_{ij}\}} & \text{for cost criterion } j\text{th} \end{cases}$$

Step 2: Determining the maximum value M_i among the criteria as illustrated in Eq. (17). Consider M_i be achieved on k_i number of criteria ($k_i \leq n$), say, $C_{\gamma_i(1)}, \dots, C_{\gamma_i(k_i)} \in \{C_1, \dots, C_n\}$; set $J_i = \{\gamma_i(1), \dots, \gamma_i(k_i)\}$.

$$M_i = \max_{j \in J_i} \{\tilde{r}_{ij}\} \quad (17)$$

Step 3: Determining the preference index using Eq. (18) defined as $P_j(A_i, A_{i'})$ where $i, i' \in I, i \neq i'$.

$$P_j(A_i, A_{i'}) = \begin{cases} 0 & \text{if } \tilde{r}_{ij} \leq \tilde{r}_{i'j} \\ \tilde{r}_{ij} - \tilde{r}_{i'j} & \text{otherwise} \end{cases} \quad (18)$$

Step 4: Calculating the cumulative preference index using Eq. (19) defined as $\pi(A_i, A_{i'})$ where $i, i' \in I, i \neq i'$.

$$\pi(A_i, A_{i'}) = \frac{1}{k_i} \sum_{j \in J_i} P_j(A_i, A_{i'}) \quad (19)$$

Step 5: Determining the leaving (positive) and entering (negative) outranking flows using Eqs. (20) and (21) respectively. Then, calculate the net outranking flow using Eq. (22).

$$\phi^+(A_i) = \frac{1}{m-1} \sum_{\substack{i'=1 \\ i \neq i'}}^m \pi(A_i, A_{i'}), i \in I \quad (20)$$

$$\phi^-(A_i) = \frac{1}{m-1} \sum_{\substack{i'=1 \\ i \neq i'}}^m \pi(A_{i'}, A_i), i \in I \quad (21)$$

$$\phi(A_i) = \phi^+(A_i) - \phi^-(A_i), i \in I \quad (22)$$

Step 6: Ranking the alternatives by considering the largest value of $\phi(A_i)$ as the best alternative.

3. Results and discussion

This section presents results for the proposed hybrid system through simulation in HOMER Pro. Considering six scenarios proposed in this study, the best scenario was selected by utilizing the CRITIC-PROMETHEE approach.

The scenarios' results were categorized into economic, technical, and environmental practicalities as the main criteria in the MCDM. Under economic practicalities, net present value (NPC), leveled cost of electricity (LCOE), Initial Capital (IC), and Operation cost (OpC) were considered for each scenario. Similarly, technical practicalities included Energy Production (EP), Unmet Electric Load (UL), Excess Electricity (EE), and Capacity Shortage (CS). Environmental practicalities included emissions of Carbon Dioxide, Carbon Monoxide, Unburned Hydrocarbons, Particulate Matter, Sulfur Dioxide, and Nitrogen oxide. Table 4 gives the summary of simulated results for all scenarios.

3.1. Selection of the best scenario using MCDM

3.1.1. CRITIC approach

For economic practicalities, a large value of NPC was considered the best scenario while for LCOE, initial capital, and operation cost, small values were considered the worst scenario. For technical practicalities,

Table 4
Summary of simulated results for six scenarios.

	Scenario 1	Scenario 2	Scenario 3	Scenario 4	Scenario 5	Scenario 6
Economic results						
NPC (\$)	116,853	104,064	79,511	86,099	88,954	118,475
LCOE (\$/kWh)	0.1397	0.1244	0.09508	0.103	0.1064	0.1428
Initial capital (\$)	9,500	24,137	24,319	29,858	10,000	55,949
Operation cost (\$)	7752.3	5771.75	3985.9	4061.33	5701.53	4515.19
Technical results						
Energy production (kWh/yr)	60,502	65,843	66,973	67,012	61,581	101,465
Unmet electric load (%)	0	0	0	0	0	494
Excess electricity (%)	116	4447	5611	5717	1195	39987
Capacity shortage (%)	0	0	0	0	0	664
Environmental results						
Carbon dioxide (kg/yr)	39191.00	27067.00	20356.00	20287.00	29040.00	20629.00
Carbon monoxide (kg/yr)	16.00	14.40	0.02	0.02	0.03	0.00
Unburned hydrocarbons (kg/yr)	0.70	0.63	0.00	0.00	0.00	0.00
Particulate matter (kg/yr)	0.10	0.09	0.00	0.00	0.00	0.00
Sulfur dioxide (kg/yr)	165.00	113.00	88.20	87.90	126.00	89.40
Nitrogen oxides (kg/yr)	92.80	66.10	43.10	43.00	61.50	43.70

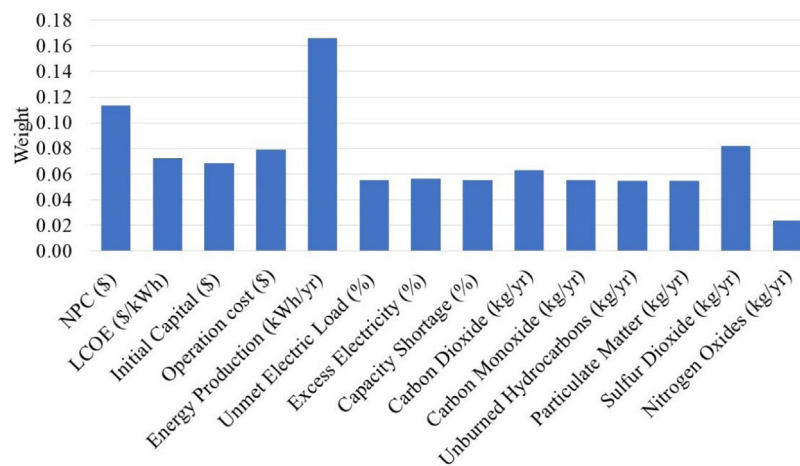


Fig. 5. Calculated weights of all sub-criteria using the CRITIC approach.

the large value of energy production was considered the best scenario while unmet electric load, excess electricity, and capacity shortage, small values were considered the worst scenario. For all environmental practicalities, small values were considered the best scenario. This study considered equal weights of economic, technical, and environmental practicalities, the calculated weights of all sub-criteria using the CRITIC approach are presented in Fig. 5. The results gave high weight to energy production, NPC, sulfur dioxide, and operation cost respectively.

3.1.2. PROMETHEE II

The weights from the CRITIC approach were used as input to the PROMETHEE II approach. In this study, NPC and energy production were considered beneficial criteria while the rest of the criteria were regarded as non-beneficial criteria. Table 5 gives the summary of the ranking of all scenarios using their net outranking. The results show that the most suitable scenario is Scenario 3 seconded by Scenario 4. Scenario 3 is a hybrid system comprised of a grid, biogas, and solar PV without a storage unit. On the other hand, Scenario 1 is the worst Scenario which is comprised of a grid and diesel hybrid system.

3.1.3. Sensitivity analysis

In this study, sensitivity analysis is used to validate the MCDM results and the robustness of the technique. The primary goal of this sensitivity analysis was to determine the new ranking of the Scenarios using their net outranking values. This was done by varying the initial

Table 5

Summary of ranking of all scenarios using their net outranking.

	(+) Entering flow	(-) Entering flow	Net outranking	Rank
Scenario 1	0.131	0.426	-0.296	6
Scenario 2	0.133	0.252	-0.119	5
Scenario 3	0.253	0.108	0.145	1
Scenario 4	0.241	0.098	0.143	2
Scenario 5	0.189	0.161	0.028	4
Scenario 6	0.364	0.266	0.099	3

weights of the categories in different cases and comparing them to the current case. All categories were given equal weights (33.3%) in Case S1, the original Case. For Case S2, the economic and technical categories were given 35% each, while the environmental category was given 30%. For Case S3, the economic, technical, and environmental categories were assigned weights of 40%, 35%, and 25%, respectively. For Case S4, the economic, technical, and environmental categories were assigned weights of 50%, 25%, and 25%, respectively. For Case S5, the economic, technical, and environmental categories were assigned weights of 25%, 50%, and 25%, respectively. For Case S6, the economic, technical, and environmental categories were assigned weights of 25%, 25%, and 50%, respectively. The sensitivity results in this study revealed that the results in this study were consistent and robust. Though there are some fluctuations in the net outranking of several Cases, as shown in Fig. 6, most of them were consistent with the

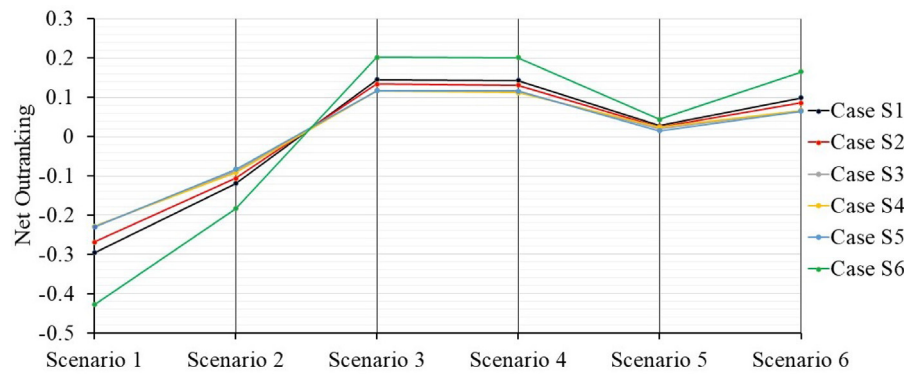


Fig. 6. Sensitivity results for all scenarios using six Cases.

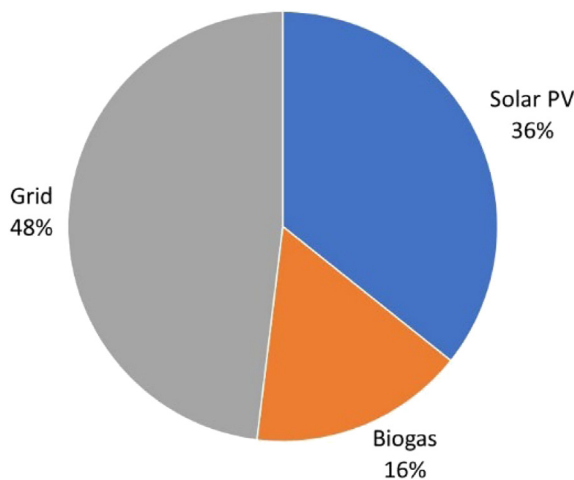


Fig. 7. Annual energy generation share for the proposed system.

current case (Case S1). Furthermore, the ranking scenarios remained constant in all cases.

3.2. Analysis of the best scenario

Six scenarios were simulated using HOMER pro and the results were further analyzed using MCDM approaches, CRITIC-PROMETHEE II. The MCDM approaches utilized economic, technical, and environmental categories. The results showed that Scenario 3 is the best which is comprised of the grid, biogas, and solar PV without a storage unit. The winning system architecture has grid capacity, a generator of 20 kW capacity of electricity from biogas, a 13.1 kW capacity of solar PV, and a system converter of 9.19 kW. The system has no battery storage which may be beneficial considering the lifetime of the battery.

The results had shown that the AC load consumption for the whole year is 60,386 kWh. The annual contribution of solar PV is 23,917 kWh, the biogas generator is 10,850 kWh and the grid is 32,184 kWh. Fig. 7 shows the share of each system component towards the hybrid system. The results illustrate that a huge percentage of electricity is obtained from the grid with a 48% contribution, however, the combined contribution of solar PV and biogas is more than that of the grid. This microgrid requires 165 kWh/day and has a peak of 17 kW. In the proposed system, the monthly generation sources satisfying the load are illustrated in Fig. 8. The annual excess electricity generation for the system is 5,590 kWh representing 8.35% of the total generation. This excess electricity is advantageous because it can allow for additional load. After all, there is no capacity shortage or unmet electricity load in the system.

Table 6
Summary of parameters for biogas production.

Parameter	Quantity
Average human weight (kg)	50
Discharge per day (kg)	0.5
TS value of fresh discharge (% by wt)	20
Daily amount of water required (kg)	360
Number of students	480
Total discharge (kg/day)	240
TS of fresh discharge	48
Total volume of feedstock per day (m ³)	600
Gas produced per day (m ³)	22.944
Digester volume (m ³)	24

3.2.1. Biogas production

Biogas calculations were carried out to determine levels of production such as the amount of gas. Results show that human excreta can give daily gas production amounting to 22.944 m³. The system also requires about 360 liters of water per day. Table 6 shows the summary of parameters for biogas production from human excreta at Blantyre secondary school. These results can further be used in the design of a biogas digester.

3.2.2. Solar PV

The proposed solar PV rated capacity was 13.1 kW with a mean output power of 2.73 kW and 65.5 kWh/day. The annual energy production for solar PV was estimated at 65,560 kWh/year and the annual operation time was 4,376 h per year. Solar PV production occurs during daylight, which is usually between 06:00 a.m. and 6:00 p.m. as illustrated in Fig. 9. The required capital cost for a solar PV system is \$12,475.82 with an annual O&M cost of \$181.86. Solar PV system alone had a levelized cost of 0.0382 \$/kWh.

3.2.3. Biogas generator

The results suggested a 20 kW biogas generator for the system. This generator had an operation time of 995 h per year with 489 starts in a year. Thus, the generator has a lifetime of 20.1 years. The generator requires a fixed generation cost of \$1 per hour because it requires intensive labor and maintenance. The generator's electrical production is 10,850 kWh per year with a mean electrical power output of 10.9 kW. The generator has a capacity factor of 6.19% because the generator operates for a few hours a day and stays off most of the time. Fig. 10 shows the performance of the biogas generator which shows high-frequency use in evening hours between 18:00 and 00:00. The annual fuel consumption of a generator is 9.29 tons with specific fuel consumption of 2.18 kg/kWh. Thus, the estimated annual fuel energy input is 36,195 kWh/yr. This implies that comparing the input energy and output electrical energy, the mean electrical efficiency was 30%. The generator fuel price is 0.05 \$/kg with an annual generator maintenance cost of \$597. The required capital cost for the generator is \$ 10,000 and the O&M cost is \$8,267.22.

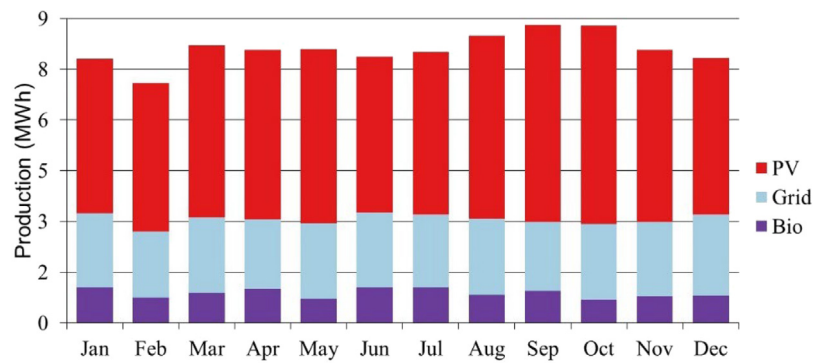


Fig. 8. Monthly electricity generation for the proposed system.

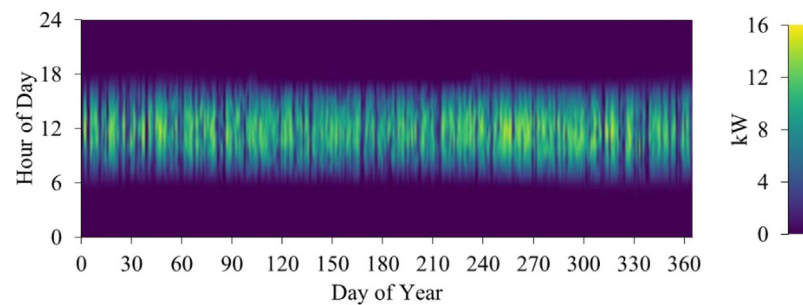


Fig. 9. Daily solar PV performance.

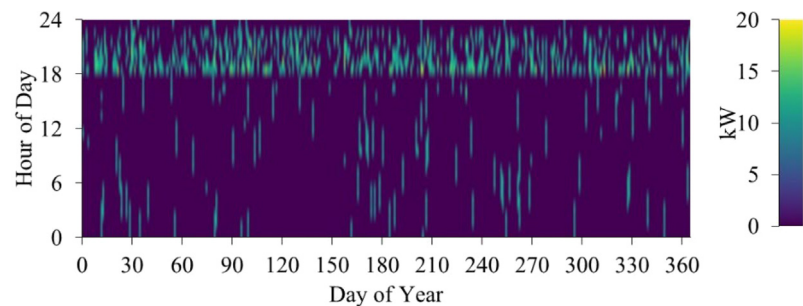


Fig. 10. Daily biogas generator performance.

3.2.4. Grid electricity

The main objective of this paper was to reduce dependence on the grid using hybrid systems. The grid was designed to be off and sometimes partially operational. The grid is usually put off during daylight because of the solar PV generation peak. The school is already connected to the grid; hence no capital cost is required. Electricity purchases from the grid throughout the project's lifetime is summing up to \$94,406.62. Table 7 shows monthly energy purchased from the grid, peak demand, and its corresponding energy charge.

3.2.5. Systems converter

The estimated capacity of a system converter was 9.19 kW. The mean output power of the converter is 2.12 kW with a maximum output power of 9.19 kW. The simulated results have shown that the capacity factor of the system is 23% with an annual operation time of 4,205 h. The annual converter's energy input and output are 19,522 kWh and 18,546 respectively representing 976 kWh of losses. The converter operations are dependent on the solar PV production which occurs during daylight, which is usually between 06:00 a.m. and 6:00 p.m. as illustrated in Fig. 11. The required capital, replacement, and O&M cost of the system's converter throughout the project lifetime was estimated as \$1,838.78, \$603.73, and \$509.27 respectively.

Table 7

Monthly energy purchased from the grid and their charge.

Month	Energy purchased (kWh)	Peak demand (kW)	Energy charge (\$)
January	2,751	10.0	275.08
February	2,485	9.95	248.50
March	2,797	9.98	279.70
April	2,510	9.99	251.04
May	2,701	9.95	270.11
June	2,713	9.99	271.34
July	2,659	9.97	265.91
August	2,778	9.98	277.82
September	2,493	9.97	249.29
October	2,681	9.93	268.13
November	2,635	9.98	263.54
December	2,979	9.99	297.93

3.2.6. Financial summary

The simulated results show that the optimal system had an initial capital cost and capital cost of \$24,315 and \$12,476 respectively. The net present cost (NPC) and operating cost of the system were estimated as \$79,511.05 and \$3,985.9 respectively. The proposed system's levelized cost of electricity (LCOE) was 0.09508 \$/kWh. From the

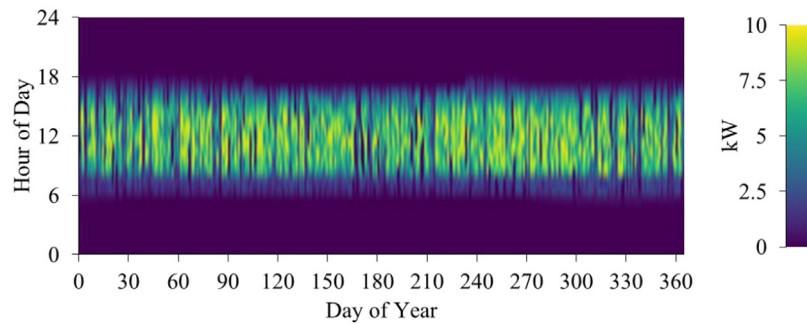


Fig. 11. Daily systems converter performance.

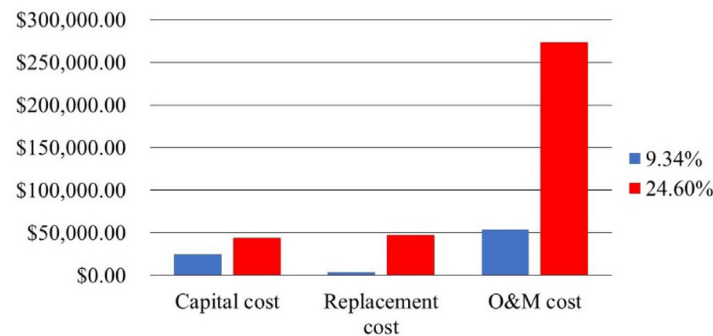


Fig. 12. Comparing inflation rates on the financial performance of the system's design.

simulated results, the designed system had a simple payback period of 9.03 years and a discounted payback of 12.49 years. The internal rate of return (IRR) was estimated at 11.1%.

3.3. Emission

Septic tanks produce polluting gases such as carbon dioxide, methane, and nitrous oxide. This makes the collection of human excreta dangerous for the environment. Additionally, sewage management requires a lot of money, resulting in a waste of money. On the other hand, biogas production reduces these gases by utilizing methane gas for energy generation [29]. However, there is a small amount of pollution during energy generation. This proposed system was estimated to release 20,356 kg/yr of carbon dioxide, 0.02 kg/yr of carbon monoxide, 88.2 kg/yr of sulfur dioxide, and 43.1 kg/yr of nitrogen oxides. Environmentally, the system can help to reduce the emission of gases into the atmosphere, thereby preventing global warming. The emissions from energy generation from biogas are relatively lower and less harmful than those from septic tanks.

3.4. Impact of inflation on the system

The system results were compared based on the 2022 inflation rate of 24.6% and the 2021 inflation rate of 9.34%. This is because the 2022 inflation rate is considered inflation due to the world crisis. This sudden inflation increase has a serious impact on financial performance. In this study, when inflation was pegged at 9.34%, financial parameters such as capital, replacement, and operation and maintenance costs were low as compared to 24.6%. Worst still, there is a serious increase in the replacement and O&M costs as illustrated in Fig. 12. Furthermore, the payback period also increases from 9.03 to 13.54 years for the inflation of 9.34% and 24.6% respectively.

3.5. Comparing the proposed design with other grid-connected systems

In this section, the proposed system is compared with other studies in different geographical locations and technical capabilities. In Iran, Jahangir et al. [74] investigated a grid-connected hybrid system comprised of solar PV, biogas, diesel, and battery. The study reported the system can reduce carbon emissions by 788 tons per year with reduced LCOE. In India, Jain & Sawle [75] investigated a grid-connected system for a town containing solar PV, wind, and hydrogen. A hybrid system in Egypt with grid, solar PV, wind, and battery reported a reduced LCOE [76]. Lemence & Tamayao [77] considered an off-grid and on-grid hybrid system for a rural healthcare facility in Philippines using HOMER pro. According to the study, solar PV, battery, and diesel systems are best for off-grid options while solar PV, grid, and PV systems are best for on-grid options. In Malawi, Chisale & Mangani [55] used HOMER pro to study a grid-connected solar and battery system for a commercial building and found that it is more affordable than a grid system alone. In France, Islam [78] used HOMER pro to optimize a solar PV grid-connected system for an office building. According to the study, emissions were reduced by 90% and LCOE was decreased by 10%. In Indonesia, Riayatsyah et al. [27] study a grid-connected system for a university. The optimized system containing solar PV and wind reported a reduction in LCOE. As shown in Table 8, the LCOE for the proposed system is relatively high when compared to most grid-connected studies. In comparison to the Philippine system, however, is low [77]. Furthermore, different studies have different RF, emphasizing that different optimal system designs are dependent on location, system combination, and other technical and economic factors.

3.6. Managerial insights

Malawi's electricity problems are critical, with little progress made in decades. Overdependence on the national grid strains the grid because it must be upgraded, and power capacity increased regularly. To

Table 8
Comparing results with other grid-connected systems.

Location	Year	System	LCOE (\$/kWh)	RF (%)	Reference
Iran	2022	Grid, PV, BG, DG, BT	0.083	66.9	[74]
India	2021	Grid, PV, WT, HG	0.034	91.5	[75]
Egypt	2020	Grid, PV, WT, BT	0.0322	62.56	[76]
Philippines	2021	Grid, PV, DG, BT	0.13	61	[77]
Malawi	2021	Grid, PV, BT	0.0372	89.5	[55]
Indonesia	2022	Grid, PV, WT, BT	0.0446	82	[27]
Malawi	–	Grid, PV, BG	0.095	46.7	Present study

address these issues, the government, energy users, and suppliers must all take on individual responsibilities. Furthermore, Government must incorporate Policies and incentives to support small-scale renewable energy generation including at the household level. Thus, households, businesses, and other electricity users must be self-sufficient or partially rely on the grid. The following are some of the benefits of such systems for small-scale applications:

- No interruptions due to grid outages
- Reduce grid pressure to make electricity available to other users.
- It might eventually result in cost savings for the institution.
- Infrastructure costs like grid extension and maintenance are eliminated from the supply side.

4. Conclusion

Hybrid renewable energy systems have proven to be more reliable and provide continuous electricity supply despite the intermittent nature of energy. In this study, a grid-connected hybrid energy system is intended to address electricity issues and lower costs in a school environment. Renewable energy resources (solar PV, wind, biogas), diesel, and battery were considered for design and simulation in a school setting. Biogas generation is mainly from human excreta.

In this paper, six scenarios involving various hybrid energy combinations with grid power were taken into consideration. The HOMER Pro modeling tool was utilized in the simulation of all scenarios. The results obtained from HOMER about the simulated hybrid systems were classified into three perspectives (Economic, Technical, and Environmental). These results were further analyzed to determine the best combination for the location using the CRITIC-PROMETHEE II approach. The study demonstrated that, for all the scenarios, the most viable scenario was Scenario 3 which had a combination of the grid, biogas, and solar PV without any storage unit. The following are the main findings of the current investigation:

- The addition of a diesel generator increased the NPC values in scenarios 1 and 2. This is due to the high cost of diesel fuel. Thus, not economically viable.
- Scenario 6 had a high operating cost due to the battery's short life span, which necessitated frequent replacement and was thus not economically feasible.
- Given the intermittent nature of solar and grid outages, including a biogas generator was a viable option with a lower LCOE of 0.095 \$/kWh. As a result, scenario 3 was a viable option.
- When the LCOE of the proposed system is compared to the cost of electricity on the national grid (0.1 \$/kWh), the bills are expected to drop by approximately 32%.
- Environmentally, scenario 4 is the best with an annual carbon dioxide emission of 20,287 kg; however, the optimal system (scenario 3) has an emission of 20,356 kg, which is relatively better when compared to other scenarios.
- Inflation can have a significant impact on the system's economic performance. Thus, in this study, an increase in the inflation rate will increase capital, replacement, and O&M costs.

The proposed system has excess energy that can be sold back to the grid via net metering, which has yet to be implemented in Malawi. The study's load profile was an average annualized electrical load profile. As a result, the future scope of this study will include actual measurement of electricity use over time while considering occupant behavior. Furthermore, to design the most acceptable system, social, political, and risk factors will be incorporated into the MCDM.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors declare that the data supporting the study's findings are included in the article and other specific data will be provided upon request.

Acknowledgment

The authors would like to thank the Malawi University of Science and Technology, Malawi for providing support in several ways for this study.

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